

Microeconomic Foundations, I: Choice and Competitive Markets

Instructor's Manual

Chapter 10: The Expenditure-Minimization Problem

Notes on the Chapter

This chapter concerns the Expenditure-Minimization Problem (also known as the Dual Consumer's Problem).

Following Chapter 9, there is little surprising that goes on. The early parts of the chapter formulate the problem and undertake basic analysis; latter parts take on the duality between utility functions and expenditure functions. I essentially follow the results of Jackson (1986a), which provide necessary and sufficient conditions for an expenditure function to arise from a continuous and locally insatiable utility function. I have found that one session (of 90 minutes or so) is adequate to discuss this material, assuming a thorough discussion of Chapter 9 precedes this; perhaps the only difficulty I encounter is that the somewhat complex plot of Section 6 needs careful explication.

Notes on the Problems

For very good students, this is a good chapter to get “down in the weeds,” asking what assumptions are needed for the various results. In this regard, I recommend problems 2 through 6. An alternative is to tackle problems 8 and 9, which (essentially) redo the theory but assuming the consumer's target level of utility is given by a bundle of goods, rather than some abstract utility level. Please note: This is not as trivial as it may at first sound; at least, my redone theory takes some twists and turns that I didn't at first anticipate.¹ A third approach is to go back to cost-minimization for a firm with a single output good, described by a production function, asking students to translate the developments of this chapter into that context. None of these problems is particularly tricky, at least, for students with strong skills in mathematical analysis. (Well, Problem 9 might be graded as tricky toward the end.) But by making several of them open-ended, I try to compel students to dig into the finer structure of the chapter.

Copyright © David M. Kreps, 2012.

¹ Although I am unaware of anyone having published this modification to the basic theory, it is too obvious not to have been done by someone. So I would be very grateful if someone could supply me with a reference, especially if, as I expect, I'm not doing this as elegantly as is possible.

Solutions to Non-Starred Problems

■ 10.2. Let $k = 2$, and let $X = \{(x_1, x_2) \in \mathbb{R}^2 : x_1 > 0, x_2 \leq \ln(x_1)\}$. This is a closed *and* convex set whose CCH is $X = \{(x_1, x_2) \in \mathbb{R}^2 : x_1 > 0\}$, which is hardly closed (in fact, it is open).

Let $k = 2$, and let $X = \{(x_1, x_2) \in \mathbb{R}^2 : x_1 \geq 0, x_2 \geq x_1/(x_1^2 - x_1 + 1)\}$. If I did this right, we are looking at points above the function $x_2 = x_1/(x_1^2 - x_1 + 1)$, which, for $x_1 > 0$, is a function that starts at $(0, 0)$, increases for awhile, and then decreases, asymptoting at ∞ to $x_2 = 0$. This is clearly a continuous function (the denominator stays bounded away from zero), so the set is closed. But because of the asymptote, the convex hull of the set is $\{(0, 0)\} \cup \{(x_1, x_2) \in \mathbb{R}^2 : x_1 \geq 0, x_2 > 0\}$, which is not closed.

■ 10.4. I'll organize my answer to this question based on the assumptions needed.

First, there are properties of $\mathbf{H}$ and e that require no assumptions about u beyond assuming that $u : \mathbb{R}_+^k \rightarrow \mathbb{R}$:

For arbitrary u , $p \rightarrow e(p, v)$ is concave, continuous, homogeneous of degree 1, and nondecreasing (for all v such that, for some x , $u(x) \geq v$). For every p , $v \rightarrow e(p, v)$ is nondecreasing (over the range of v such that, for some x , $u(x) \geq v$). If $x \in \mathbf{H}(p, v)$, then $x \in \mathbf{H}(\lambda p, v)$ for all $\lambda > 0$. For all p and v such that, for some x , $u(x) \geq v$, $e(p, v) \geq 0$. And if $v \leq u(0)$, then $e(p, v) = 0$.

For the most part, these statements all follow the proofs given in the text, although if u is not upper semi-continuous, the set $\{x \in \mathbb{R}_+^k : u(x) \geq v\}$ may not be closed, which means we aren't guaranteed a solution to the EMP; this complicates some of the proofs. So let me give a sketch of this.

First, we are restricting attention to v such that, for some x , $u(x) \geq v$. (For v such that $u(x) < v$ for all $x \in \mathbb{R}_+^k$, we could define $e(p, v) = \infty$, but then we'd need to do some special pleading. I'll avoid that.) For such v and any strictly positive p , it is clear that $e(p, v) \geq 0$; $e(p, v)$ is the infimum of a nonempty set of nonnegative numbers. Moreover, if $v \leq u(0)$, then 0 is in $\{x \in \mathbb{R}_+ : u(x) \geq v\}$, and $p \cdot 0 = 0$, hence for such v , $e(p, v) = 0$.

The statement that $x \in \mathbf{H}(p, v)$ implies $x \in \mathbf{H}(\lambda p, v)$ for all $\lambda > 0$ should need no further elaboration.

The statement that $v \rightarrow e(p, v)$ is nondecreasing follows from the general observation that the infimum over a larger set (by set inclusion) is necessarily no larger; increasing v shrinks the feasible set.

For any p and v (in the relevant range), I'll denote by $x^n(p, v)$ any point in $\mathbb{R}_+^k$ such that $u(x^n(p, v)) \geq v$ and $p \cdot x^n(p, v) \leq e(p, v) + 1/n$. Even if there is no solution to the EMP for the given p and v , there are points that come arbitrarily close to the infimum; we are just making a selection of such points here.

Then for any $p' \geq p$, $e(p', v) + 1/n \geq p' \cdot x^n(p', v) \geq p \cdot x^n(p', v) \geq e(p, v)$. Since n is

arbitrary here, we have $e(p', v) \geq e(p, v)$, and $p \rightarrow e(p, v)$ is nondecreasing in p .

For any p, p' , and $\alpha \in [0, 1]$, let p'' denote $\alpha p + (1 - \alpha)p'$. Then

$$\begin{aligned} e(p'', v) + 1/n &\geq p'' \cdot x^n(p'', v) = (\alpha p + (1 - \alpha)p') \cdot x^n(p'', v) = \\ &\alpha p \cdot x^n(p'', v) + (1 - \alpha)p' \cdot x^n(p'', v) \geq \alpha e(p, v) + (1 - \alpha)e(p', v). \end{aligned}$$

Since n is arbitrary here, $p \rightarrow e(p, v)$ is concave, and being concave on an open domain, it is continuous.

I'll leave the final step of showing that $p \rightarrow e(p, v)$ is homogeneous of degree 1 to you.

The following result needs no elaboration; the objective function is linear, and if u is quasi-concave, the feasible set is convex:

If u is quasi-concave, then $\mathbf{H}(p, v)$ is a convex set (for all p and v , with the understanding that the empty set is convex).

What about the result that $\mathbf{H}(p, v)$ is singleton if u is strictly quasi-concave? For one thing, we could only hope for "singleton or empty," as we haven't made assumptions yet guaranteeing a solution to the EMP. But matters are worse for this result, so let me wait for it.

If u is upper semi-continuous, a solution to the EMP exists for every $p \in R_{++}^k$ and v such that $u(x) \geq v$ for some x . Moreover, in this case, $e(p, v) > 0$ for all $v > u(0)$.

Upper semi-continuity of u is equivalent to the statement that, for each v , $\{x \in R_+^k : u(x) \geq v\}$ is closed. That's what we needed in the old proof of existence; the argument in the text that gives boundedness of the relevant portion of the feasible set stays the same. And once we know that solutions exist, we know that, for $v > u(0)$ and any p , $p \cdot x^* > 0$ for any $x^* \in \mathbf{H}(p, v)$.

But upper semi-continuity of u is insufficient to tell us that $u(x^*) = v$ for $x^* \in \mathbf{H}(p, v)$ (and $v \geq u(0)$). The argument given in the text on this point involved assuming (by way of contradiction) that $u(x^*) > v$ and then looking at αx^* for α close to one. Of course, αx^* is cheaper than x^* (we know $x^* \neq 0$, why?), and if u is continuous, then $u(\alpha x^*) \geq v$ if $u(x^*) > v$ and α is close to one. But with only upper semi-continuity, this argument fails. And in fact, consider for $k = 1$ the utility function

$$u(x) = \begin{cases} x, & \text{if } x < 1, \text{ and} \\ x + 1, & \text{if } x \geq 1. \end{cases}$$

Note that the utility function is (only) upper semi-continuous; the underlying preferences are continuous. Even so, for this utility function, if $v = 1.5$, $\mathbf{H}(p, v) = \{1\}$, and $u(1) = 2$. Moreover, for this utility function, $e(p, v)$ is the constant p for $v \in [1, 2]$, so with (only) upper semi-continuity, we lose the result that $e(p, v)$ is strictly increasing in v .

Now return to the uniqueness result. The proof given in the text went: Suppose distinct x and x' are in $\mathbf{H}(p, v)$. Let $x'' = 0.5x + 0.5x'$; if u is strictly quasi-concave, then $u(x'') > v$, and we then looked at $\alpha x''$, getting a contradiction. But this depends on u being continuous (or, at least, lower semi-continuous): We have

If u is strictly quasi-concave and lower semi-continuous, then for each p and v , $\mathbf{H}(p, v)$ is either singleton or empty.

Another result from the early propositions is that $e(p, v)$ is unbounded above in v for each p . The proof we gave required global insatiability—and it is clear that if the consumer in question is globally satiated at some point x^* , then $e(p, v)$ is bounded above by $p \cdot x^*$ (for all v for which it is finite)—but the proof also made use of (at least) upper semi-continuity: Suppose by way of contradiction that $e(p, v)$ is bounded above by some value K for given p , and look at the set $\{x \in R_+^k : p \cdot x \leq K\}$. If u is upper semi-continuous, there is some x^* in this set that maximizes u over the set; now apply global insatiability to the point x^* to find a feasible utility level v such that, for all x for which $u(x) \geq v$, $p \cdot x > K$.

But without upper semi-continuity, it is possible that $e(p, v)$ is bounded above. Try the utility function u for $k = 1$ given by $u(x) = x - \lfloor x \rfloor$, where $\lfloor \cdot \rfloor$ denotes the “greatest integer less than or equal to” function. These preferences are globally insatiable and, in fact, locally insatiable. The range of v for which there is x with $u(x) \geq v$ is $(-\infty, 1)$. And $e(p, v)$ for $v \in [0, 1)$ is pv . So:

If u is upper semi-continuous and globally insatiable, $e(p, v)$ is unbounded above in v .

This leaves us with the full application of Berge's Theorem; that is, results that show that $(p, v) \rightarrow e(p, v)$ is continuous and $(p, v) \Rightarrow \mathbf{H}(p, v)$ is upper semi-continuous. In Proposition 10.3, we got this result by assuming that u is continuous and locally insatiable.

If u is upper semi-continuous, then $(p, v) \rightarrow e(p, v)$ is lower semi-continuous on the domain of e .

(This is shown in the *Student's Guide* as the solution to Problem 3 in this chapter, but is repeated here for completeness.) Suppose $\{(p^n, v^n)\}$ has limit (p, v) , and $\lim_n e(p^n, v^n) < e(p, v)$. (Because I said *on the domain of e* , it is assumed that for there are some $\hat{x}^n$ and $\hat{x}$ such that $u(\hat{x}^n) \geq v^n$ and $u(\hat{x}) \geq v$, so that $e(p^n, v^n)$ and $e(p, v)$ are all well defined.) We'll derive a contradiction, which then shows lower semi-continuity. Let x^n be a solution to the EMP for p^n and v^n . (Since u is upper semi-continuous, we know that solutions exist.) Because p is strictly positive, $\lim_n p^n = p$, and, eventually, $p^n \cdot x^n = e(p^n, v^n) < e(p, v)$, one can find a bounded set that contains this sequence, and so by looking along a subsequence if necessary, we can assume the sequence converges to some x^* . Since $u(x^n) \geq v^n$ and u is upper semi-continuous, we know that $u(x^*) \geq \lim_n v^n = v$. Therefore, x^* is feasible for the EMP at (p, v) , and $e(p, v) \leq p \cdot x^*$. But by continuity of the dot product, $p \cdot x^* = \lim_n p^n \cdot x^n = \lim_n e(p^n, v^n)$. That does it.

The next result we give shows that for upper semi-continuity of $e(p, v)$, continuity of u is unimportant. Instead, it is local insatiability that is key.

If u is locally insatiable, then $(p, v) \rightarrow e(p, v)$ is upper semi-continuous.

Suppose $\{(p^n, v^n)\}$ has limit (p, v) and $\lim_n e(p^n, v^n) > e(p, v)$. For any $\epsilon > 0$, let x^ϵ be such that $u(x^\epsilon) \geq v$ and $p \cdot x^\epsilon \leq e(p, v) + \epsilon$. Since u is locally insatiable, we can find a sequence $\{x^\ell\}$ with $\lim_\ell x^\ell = x^\epsilon$ and $u(x^\ell) > u(x^\epsilon)$ for all ℓ . We will now create a subsequence of the n sequence: Let n_1 be the smallest index such that $u(x^{n_1}) \geq v^{n_1}$. Let n_2 be the smallest index greater than n_1 such that $u(x^{n_2}) \geq v^{n_2}$. And so forth; since the v^n converge to v and each $u(x^\ell) > u(x^\epsilon)$, no matter how far out in the n sequence you are, you can find an n further out such that v^n is less or equal the next $u(x^\ell)$ to come along. But then along this subsequence, x^ℓ is feasible for the EMP at (p^{n_ℓ}, v^{n_ℓ}) , and so

$$e(p^{n_\ell}, v^{n_\ell}) \leq p^{n_\ell} \cdot x^\ell.$$

And as ℓ goes to infinity, the right-hand side of this inequality converges to $p \cdot x^\epsilon$. Hence

$$\lim_\ell e(p^{n_\ell}, v^{n_\ell}) \leq p \cdot x^\epsilon \leq e(p, v) + \epsilon.$$

Since $\epsilon > 0$ is arbitrary here,

$$\lim_\ell e(p^{n_\ell}, v^{n_\ell}) \leq e(p, v),$$

which contradicts the original supposition of this paragraph.

To round out this result, consider for $k = 1$ the utility function

$$u(x) = \begin{cases} x, & \text{if } x < 1, \\ 1, & \text{if } x \in [0, 2], \text{ and} \\ x - 1, & \text{if } x > 2. \end{cases}$$

That is, preference rises monotonically, except for the interval $[0, 2]$ over which it is flat. This utility function is both continuous and nondecreasing, but it is not (of course) locally insatiable. And it gives the expenditure function

$$e(p, v) = \begin{cases} pv, & \text{if } v \leq 1, \text{ and} \\ p(v + 1), & \text{if } v > 1. \end{cases}$$

This is, of course, lower semi-continuous.

Putting the two results together, to show that the expenditure function is jointly continuous in p and v , we need that u is upper semi-continuous and locally insatiable. Is this enough to derive the full power of Berge's Theorem? It is:

If u is locally insatiable, then $(p, v) \Rightarrow \{x \in R_+^k : u(x) \geq v\}$ is lower semi-continuous. Moreover, if u is (in addition) upper semi-continuous, the other requirements of Berge's Theorem are met, and $(p, v) \Rightarrow \mathbf{H}(p, v)$ is upper semi-continuous.

I refer you to the proof in the textbook of Proposition 10.3. The argument that the original constraint correspondence is lower semi-continuous makes no use of continuity properties of u ; it only uses local insatiability of u to produce the sequence $\{y^m\}$, which is then used to show lower semi-continuity. Of course, the objective function is continuous. And careful examination of the proof will show that we never use continuity of u beyond upper semi-continuity.

■ 10.5. This is Problem 2.6 once again, which is proved in the *Guide*, but to repeat the proof (in slightly different terms):

Suppose u is continuous, nondecreasing, and locally insatiable. For any x and x' such that x' is strictly greater than x in each component, let $a = \min\{x'_i - x_i; i = 1, \dots, k\}$; that is, a is the smallest difference in corresponding components between x' and x . By assumption, $a > 0$. Since u is locally insatiable, there is some x'' within $a/2$ of x that is strictly preferred to x or, in u terminology, that has $u(x'') > u(x)$. Let x^0 be the join of x'' and x ; that is, $x_i^0 = \max\{x''_i, x_i\}$ for $i = 1, \dots, k$. Since u is nondecreasing, $u(x^0) \geq u(x'') > u(x)$. But by the triangle inequality, x''_i is within $a/2$ of x_i for each i , so the same is true of x_i^0 , and so $x' \geq x^0$. This then implies that $u(x') \geq u(x^0) > u(x)$. The function u is strictly increasing for strict increases in its argument.

Conversely, suppose that u is continuous and strictly increasing for strict increases. Then for any x and $\epsilon > 0$, $x' = x + (\epsilon/k, \epsilon/k, \dots, \epsilon/k)$ is within ϵ of x and is a strict increase of x , so $u(x') > u(x)$. This shows local insatiability.

Finally, suppose we have $x' \geq x$. Let $x^n = x' + (1/n, 1/n, \dots, 1/n)$. Then x^n is a strict increase on x , hence $u(x^n) > u(x)$. But as $n \rightarrow \infty$, $x^n \rightarrow x'$, so by continuity of u , $u(x') = \lim_n u(x^n) \geq u(x)$ —passing to the limit preserves the weak (not the strict) inequality. This shows that u is nondecreasing.

■ 10.7. Suppose the underlying utility function is continuous and locally insatiable. Then good i is normal if and only if it is Hicks-normal.

In one direction: Suppose good i is normal. Fix prices p and utility levels v and v' with $v > v' \geq u(0)$. We know that $\mathbf{D}(p, e(p, v)) = \mathbf{H}(p, v)$ and $\mathbf{D}(p, e(p, v')) = \mathbf{H}(p, v')$, and we know that e is strictly increasing in its second argument, so that $e(p, v) > e(p, v')$. But then if $h \in H(p, v)$ and $h' \in H(p, v')$, it follows that $h \in \mathbf{D}(p, e(p, v))$ and $h' \in \mathbf{D}(p, e(p, v'))$, and so $h_i \geq h'_i$ since i is normal. This then implies that i is Hicks-normal.

For the other direction, we first must show that $\nu(p, y)$ is strictly increasing in y for every p . This is proved in the text next chapter (in Proposition 11.4), but it is straightforward so to give a proof here: Fix $p \in R_{++}^k$ and $y' > y$. Since u is continuous, there is some x' that solves the CP at p and y' . Of course, x' is feasible if prices are p and

income is y . We know from Proposition 3.1d that every solution of the CP at p and y must satisfy $p \cdot x = y$, since u is locally insatiable. Hence, since $p \cdot x' = y' < y$, x' cannot be a solution at p and y : $v(p, y) > u(x') = v(p, y')$. And with this, the proof is a marshalling of definitions and application of Proposition 10.18 similar to the preceding paragraph.

■ 10.9. The problem asks you to develop a theory for the modified expenditure function, but it will be helpful to carry along the set of solutions $\mathbf{H}(p, x^0) = \arg \min\{p \cdot x : u(x) \geq u(x^0), x \in R_{++}^k\}$. Also, we use MEMP to abbreviate the *modified expenditure-minimization problem* and $\text{MEMP}(p, x^0)$ to denote the problem at the parameters p and x^0 . We always assume that u is continuous, nondecreasing, quasi-concave, and locally insatiable; recall that the conjunction of continuous, nondecreasing, and locally insatiable imply that it is strictly increasing for strict increases. The following preliminaries are all obvious extensions/modifications of the similar results in the chapter:

- a. A solution to the $\text{MEMP}(p, x^0)$ exists for all p and x^0 . If $x \in \mathbf{H}(p, x^0)$, then $x \in \mathbf{H}(\lambda p, x^0)$. The set $\mathbf{H}(p, x^0)$ is convex for all p and x^0 . If $x \in \mathbf{H}(p, x^0)$, then $u(x) = u(x^0)$.

Next comes Berge's Theorem.

- b. The function $(p, x^0) \rightarrow e(p, x^0)$ is continuous and the correspondence (p, x^0) to $\mathbf{H}(p, x^0)$ is upper semi-continuous.

This may not seem quite so straightforward, but it is still a relatively direct extension of Proposition 10.3. Next we have

- c. The expenditure function is homogeneous of degree 1 in p , for x^0 fixed. It is nonnegative valued, and it satisfies $e(p, 0) = 0$. (The first zero here is the origin in R_+^k .) It is nondecreasing in both p and x^0 . It is concave in p for fixed x^0 , and it is unbounded in x^0 for each p .

These are all relatively straightforward extensions of similar results. But note, the standard expenditure function is strictly increasing in v ; here we only know that the modified expenditure function is nondecreasing in x^0 . (And we know that it is nondecreasing only because we assume u is nondecreasing.) But while it is nondecreasing (only) in x^0 , we expect it to be strictly increasing in $u(x^0)$ for each fixed p . And, of course, it is: If $u(x^0) > u(x^1)$ and, for fixed p , x^* solves the $\text{MEMP}(p, x^0)$, then we know that $u(x^*) = u(x^0)$. But then by continuity of u , αx^* for α close to one is feasible for the $\text{MEMP}(p, x^1)$, and it costs strictly less than x^* at p . (We know $x^* \neq 0$ because $u(x^*) = u(x^0) > u(x^1) \geq u(0)$.) Follow this logic out to the end, and we have the following:

- d. For all x^0 and x^1 from R_+^k , either $e(p, x^0) > e(p, x^1)$ for all p or $e(p, x^0) = e(p, x^1)$ for all p or $e(p, x^0) < e(p, x^1)$ for all p . Moreover, if the first case holds, $x^1 \not\geq x^0$, and symmetrically for the third case.

Because we are assuming that u is nondecreasing, continuous, and locally insatiable, we know (see problem 5) that u is strictly increasing for strict increases. So we can

continue result d as follows:

d continued. Moreover, if x is strictly larger than x' in all components, then $e(p, x) > e(p, x')$ for all p .

We can paraphrase d as follows: For each and every p , $x \rightarrow e(p, x)$ is an equivalent representation of the preferences given by u . Of course, we know that every representation of a convex preference relation is quasi-concave, and so:

e. For fixed p , $x \rightarrow e(p, x)$ is quasi-concave.

There is no analogue to this in the case of the standard expenditure function, because there the "second" argument is v , e is strictly increasing in v , and (of course) strictly increasing functions are always strictly quasi-concave.

Going back to our mimicry of developments in the text, we want to use the concavity of e in p and discuss its supergradients. Fix some x^0 , and look at $p \rightarrow e(p, x^0)$. This is concave, and all p s are interior, so there is at least one supergradient of e in p at each p (for this fixed x^0). Moreover, I assert that (just as in the standard theory) every supergradient has the form $p \cdot x^*$ for some $x^* \in R_+^k$. To reiterate the argument:

Since e is homogeneous of degree 1 in p , Proposition A3.26 ensures that any supergradient has the form of scalar multiplication times some vector x^* . Suppose x^* had a negative component, say in position i . Let $\hat{p}$ be the price vector with M in coordinate position i and 1 everywhere else, so that $\hat{p} \cdot x^* = Mx_i^* + \sum_{j \neq i} x_j^*$. For large enough M , if $x_i^* < 0$, this is negative. But it must be $\geq e(\hat{p}, x^0)$, which is nonnegative. So x^* cannot have any negative components; $x^* \in R_+^k$. Note that this argument depended only on $p \rightarrow e(p, x^0)$ being nonnegative valued, concave, and homogeneous of degree 1.

Then:

e. If $x^ \in \mathbf{H}(p^0, x^0)$, then $p^0 \cdot x^* = e(p^0, x^0)$, while $p \cdot x^* \geq e(p, x^0)$ for all (other) p , and so x^* is a supergradient of e in p at (p^0, x^0) . Conversely, if x^* is a supergradient of e in p at (p^0, x^0) , then $x^* \in \mathbf{H}(p^0, x^0)$ and (so) $u(x^*) = u(x^0)$.*

The first half of this is easy: If $x^* \in \mathbf{H}(p^0, x^0)$, then $p^0 \cdot x^* = e(p^0, x^0)$ and $u(x^*) = u(x^0)$. Because x^* is feasible for the MEMP(p, x^0), $e(p, x^0) \leq p \cdot x^*$. That's what it takes for x^* to be a supergradient of e in p at (p^0, x^0) .

Conversely, suppose x^* is a supergradient of e in p at (p^0, x^0) , so that $p^0 \cdot x^* = e(p^0, x^0)$ and $p \cdot x^* \geq e(p, x^0)$ for all p . If we know that $u(x^*) \geq u(x^0)$, then $p^0 \cdot x^* = e(p^0, x^0)$ ensures that $x^* \in \mathbf{H}(p^0, x^0)$. So, do we know that $u(x^*) \geq u(x^0)$? Suppose not. Because u is continuous, this means that the point x^* is disjoint from the closed convex set $\{x \in R_+^k : u(x) \geq u(x^0)\}$. Use the Strict-Separation Theorem to produce a nonzero price vector $\hat{p}$ and constant b such that $\hat{p} \cdot x^* < b < \hat{p} \cdot x$ for all x such that $u(x) \geq u(x^0)$. Since the set of x is comprehensive upwards (since u is nondecreasing), $\hat{p} \geq 0$ by the usual argument. And then, by the usual perturbation argument, we can add a small strictly positive vector to $\hat{p}$ and keep the inequalities; we can therefore assume that $\hat{p}$

is strictly positive. But if $b < \hat{p} \cdot x$ for all x such that $u(x) \geq u(x^0)$, we know that $e(\hat{p}, x^0) \geq b$. (In fact, we even have a strict inequality.) Hence $\hat{p} \cdot x^* < e(\hat{p}, x^0)$, a contradiction (to the supposition that x^* is a supergradient).

And, once we know that $x^* \in \mathbf{H}(p^0, x^0)$, we know that $u(x^*) = u(x^0)$. Note that as an additional corollary, we also know that $e(p, x^0) = e(p, x^*)$ for all p .

This brings us to the developments of section 5 in the text, and it is here that matters begin to take a distinctly different course. Much of the section in the text is devoted to comparisons between a utility function and its nondecreasing and quasi-concave “expenditure equivalent”; for each u , the function $\hat{u}$ created in Proposition 10.13. And a central result is (roughly) that two utility functions u and u' generate the same expenditure function if and only if they share the same CCH-level sets.

Things are going to be different here. For one thing, we’re restricting attention to utility functions that are nondecreasing and quasi-concave. But, perhaps more significantly, in this development, an expenditure function is determined not by a utility representation u of underlying preferences $\succeq$ but instead by those preferences themselves. Put differently, suppose u and u' are two utility functions (both continuous, nondecreasing, quasi-concave, and locally insatiable), both of which represent $\succeq$, even though for specific numbers v the sets

$$\{x \in R_+^k : u(x) \geq v\} \quad \text{and} \quad \{x \in R_+^k : u'(x) \geq v\}$$

are typically different. Suppose, for instance, that $u' \equiv u + 3$ or, if you want to insist on utility functions taking on nonnegative values only, $u' \equiv 3u$. From the perspective of the standard EMP as analyzed in this chapter, u and u' will produce different expenditure functions. This, frankly, is a weakness of the theory of the chapter; the object being studied (the expenditure function) changes based on *economically* meaningless changes (in the representing utility function).

Here, if u and u' represent the same preference ordering on R_+^k , they necessarily produce the same expenditure function, because

$$\{x : u(x) \geq u(x^0)\} \quad \text{is, in the end,} \quad \{x : x \succeq x^0\}.$$

Therefore, if we want a result along the line *the utility function u and u' produce different expenditure functions if and only if . . .* the second part of this will have to refer to the underlying preferences. And, in fact:

- f. The utility function u and u' produce different expenditure functions if and only if they represent different preference orderings on R_+^k ; that is, if and only if, for some pair x and x' , $u(x) \geq u(x')$ but $u'(x') > u'(x)$.*

The proof is very simple (no need for separating hyperplanes, because we’ve assumed that the u ’s are nondecreasing and quasi-concave): For any p , if $u(x) \geq u(x')$, then we

know that $e_u(p, x) \geq e_u(p, x')$, where the subscript u identifies the expenditure function generated from u . And $u'(x') > u'(x)$ implies that, for any p , $e_{u'}(p, x') > e_{u'}(p, x)$. This means that e_u cannot be the same as $e_{u'}$. (This shows that if u and u' give different orders on R_+^k , they give different expenditure functions. It should be obvious that they give the same expenditure functions if they give the same ordering, which establishes *if and only if*.)

Finally, we come to the two big results of the chapter: If we know that e was generated by some u (and, clearly at this point, we really mean, some preference ordering $\succeq$), can we recover u (or, really, $\succeq$)? And when is a candidate e an actual e ? The first question is easily answered:

g. Suppose e is the expenditure function generated by some u . Then, for every x^0 , the set $\{x \in R_+^k : u(x) \geq u(x^0)\} = \{x \in R_+^k : x \succeq x^0\}$ is precisely the set

$$\{x \in R_+^k : p \cdot x \geq e(p, x^0) \text{ for all } p \in R_{++}^k\} = \{x \in R_+^k : e(p, x) \geq e(p, x^0) \text{ for some } p\}.$$

That the second set in the display is the set of x that are weakly preferred to x^0 (and that this set is the same for any p) follows immediately from d above. The more interesting characterization is the first one displayed.

In one direction, it is clear: If $u(x) \geq u(x^0)$, then x is in the set over which $p \cdot x$ is minimized in computing $e(p, x^0)$, and so $p \cdot x \geq e(p, x^0)$ for all p is obvious. For the other direction, suppose $u(x) < u(x^0)$. Then x is not in the closed and convex set $\{x' \in R_+^k : u(x') \geq u(x^0)\}$, so a strictly separating hyperplane can be passed between them. Since the set is comprehensive upwards, the vector p that defines the hyperplane must be nonnegative (where we are orienting things so that $p \cdot x < b < p \cdot x'$ for all x' in the set). The usual small perturbation argument can be given so that we can assume p is strictly positive, and then $p \cdot x < e(p, x^0)$.

The grand finale is a result along the following lines: If $e : R_{++}^k \times R_+^k \rightarrow [0, \infty)$ satisfies the following list of conditions [list follows], then e is the (modified) expenditure function for some continuous, nondecreasing, quasi-concave, and locally insatiable utility function u , meaning that

$$e(p, x^0) = \min \{p \cdot x : u(x) \geq u(x^0), x \in R_+^k\}. \quad (\star)$$

We have a pretty good list of necessary conditions on the function e . It must be jointly continuous, nondecreasing in both arguments and strictly increasing for strict increases in the second argument, homogeneous of degree 1 and concave in its first argument, quasi-concave and unbounded above in its second argument, and it must satisfy the various conditions enumerated in d given previously.

The conditions in d are, moreover, very powerful, when combined with the other necessary conditions. The conditions in d tell us that, in the underlying preferences that

give us e (if any do), $x \succeq x'$ if and only if $e(p, x) \geq e(p, x')$ for all p . So if we choose an arbitrary p and define $u : R_+^k \rightarrow [0, \infty)$ by $u(x) = e(p, x)$, this must be a utility function that generates e . Automatically, u is continuous, nondecreasing, strictly increasing for strict increases (hence, locally insatiable), and quasi-concave. But is it true that $(\star)$ holds if e satisfies the list of conditions just given?

It is not true; we need more. The problem is one of the scale of the function e . For instance, it is obvious that for all p and x , $p \cdot x \geq e(p, x)$ is required; the cheapest way to achieve the utility of bundle x can't be any larger than $p \cdot x$. So we need to add this (certainly necessary) condition to our list. And even that isn't enough. It gives us the right upper bound on e , but e can be too small to recreate itself via $(\star)$. The result is:

h. Suppose $e : R_{++}^k \times R_+^k \rightarrow [0, \infty)$ satisfies the following list of conditions: It is jointly continuous, nondecreasing in both arguments and strictly increasing for strict increases in the second argument, homogeneous of degree 1 and concave in its first argument, quasi-concave and unbounded above in its second argument, satisfies the various conditions enumerated in d, satisfies $p \cdot x \geq e(p, x)$ for all p and x , and satisfies: If x is a supergradient of e in p at (p^0, x^0) , then $p^0 \cdot x = e(p^0, x^0) = e(p^0, x)$. Then for every p and x^0 , the equation $(\star)$ holds, for the utility function $u(x) := e(\hat{p}, x)$ (for arbitrary $\hat{p}$). Note that condition d tells us that the arbitrary choice of $\hat{p}$ is irrelevant, and we know that u is nondecreasing, continuous, locally insatiable, and quasi-concave.

With all the preparation we've done, the proof is easy.

Fix some p and x^0 and look at $\min \{p \cdot x : u(x) \geq u(x^0), x \in R_+^k\}$. The constraint set is $\{x : e(\hat{p}, x) \geq e(\hat{p}, x^0), x \in R_+^k\}$ which, by condition d, is the same as the constraint set $\{x : e(p, x) \geq e(p, x^0), x \in R_+^k\}$. By assumption, $p \cdot x \geq e(p, x)$ for all p and x , so for any x inside the constraint set, $e(p, x) \geq e(p, x^0)$, and so the minimum can be no smaller than $e(p, x^0)$.

Now let $\hat{x}$ be any supergradient of e in p at (p, x^0) . (We know that at least one supergradient exists.) That is, $p \cdot \hat{x} = e(p, x^0)$. The final condition is that $e(p, \hat{x}) = e(p, x^0)$, so $\hat{x}$ is within the set over which the minimum is taken, and so the minimum can be no larger than $e(p, x^0)$. Combining this result with the previous one gives $(\star)$.